Adaa

every outfit tells your adaa

Adaa: Project Report

Every outfit tells your Adaa

A personal wardrobe stylist that builds complete outfits only from the clothes a person already owns, with Indian traditional, Western and Indo-Western styling.

Author: Swarangi Kothawade

Live app: adaa-every-outfit-tells-your-adaa.eastasia.cloudapp.azure.com

Google Play: coming soon

Report date: November 2026

Contents

1. Problem statement
2. Proposed solution
3. Architecture
4. Workflow
5. Technology stack
6. Features
7. Feature-wise implementation
8. Security
9. Future scope

1 Problem Statement

Problem	What it looks like in daily life
"Nothing to wear"	People own many clothes but forget what they have, so they repeat the same few outfits.
Wasted money	New clothes are bought that don't go with anything already in the wardrobe.
Generic apps	Most fashion apps suggest clothes the user doesn't own, or push shopping.
Indian wear ignored	Sarees, kurtas, lehengas, jhumkas, wedding functions and festival colours are rarely understood.
Untrustworthy AI	AI image tools can "invent" clothes or show pieces that don't look like the real ones.

2 Proposed Solution

Adaa is a web app (it also installs on a phone's home screen) that works like a personal stylist for **your own** wardrobe.

Idea	How Adaa does it
Start from what you own	The user adds a photo and a few details of each piece once (clothes, jewellery, shoes, bags).
Ask naturally	Type or speak a request in English or Hinglish, e.g. "*Aaj college ke liye kuch comfy*", or tap options.
Complete, honest looks	Up to 4 different outfits, each with the reasons it works and honest cautions. If a piece is missing, Adaa says so instead of inventing it.
See the real pieces	Each look is shown as an "outfit board" made from the user's own photos with the backgrounds removed. No AI-generated images.
Indian occasions built in	Wedding functions (haldi, mehendi, sangeet…), about 30 festivals with their colours, and Hinglish words.
Wear more, buy less	Wear stats, cost per wear, and "Shop Smart" ideas for the few new pieces that would unlock the most outfits.

> **Core rule:** Adaa never invents a piece. Every outfit uses only items the user has added.

3 Architecture

Adaa has three main parts: the **app in the browser** (what the user sees), the **server** (stores data and chooses outfits), and a few **outside services** used only for specific jobs.

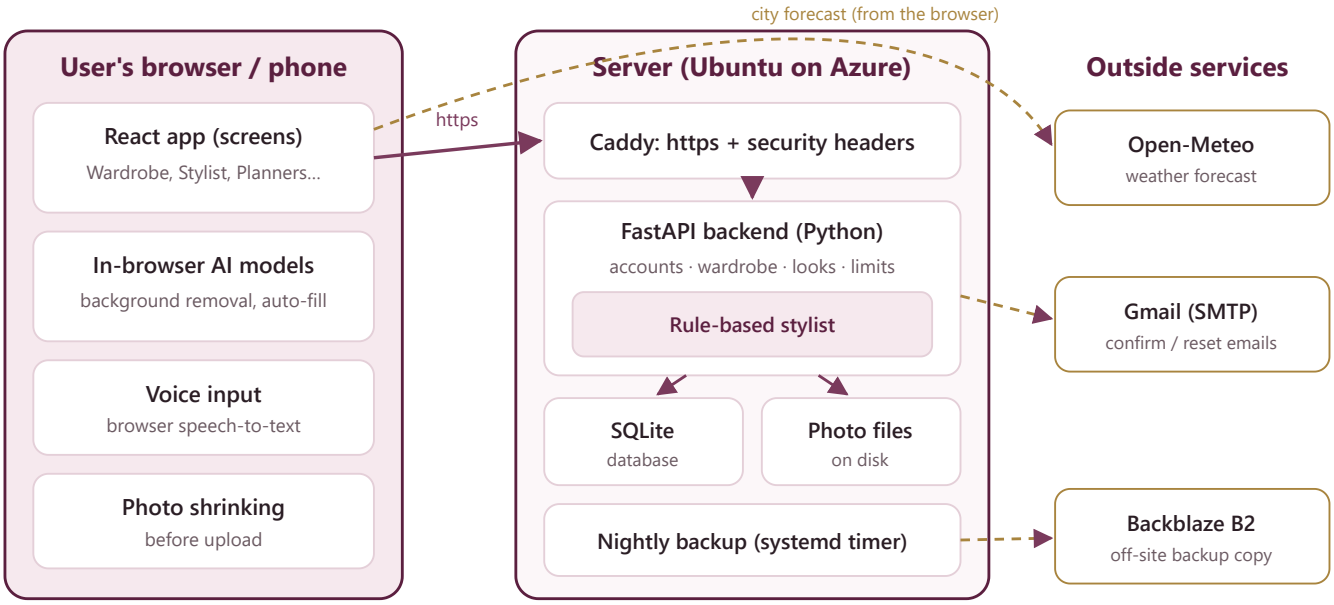
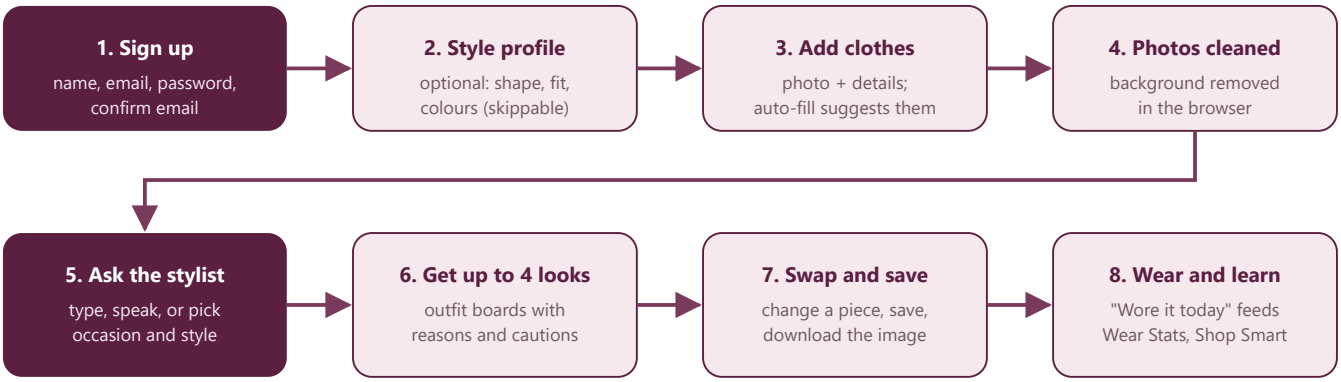


Figure 1. Adaa's architecture. Solid arrows: the app's own traffic. Dashed arrows: outside services.

Part	Job
Browser app	Shows every screen. Runs the photo-cleaning AI models on the user's own device, so wardrobe photos are never sent to an AI company. Voice uses the browser's speech-to-text (in Chrome, Google's speech service).
Caddy	The "front door": gives the site its padlock (*https*, encrypted connection) and adds security headers.
FastAPI backend	The *API* (the set of addresses the app talks to): accounts, wardrobe, looks, planners. Also serves the app's files, so only one program runs.
Rule-based stylist	Plain Python rules that pick and explain outfits (section 7).
SQLite + photo files	All data in one database file; photos as files with random names.
Backups	Every night: a copy of the database and photos, the newest 7 kept on the server, plus an off-site copy in Backblaze B2.

4 Workflow

How a person uses Adaa, from first visit to everyday use:



Planners (Wedding, Trip) and Festival styling follow the same path from step 5.

Figure 2. The user's journey in Adaa.

What happens behind one request

Step	What happens
1. Understand	The request text is read by keyword rules: occasion, style, colours, comfort, festival, and any piece the user names ("my purple navari").
2. Build	Every sensible combination is built from the user's own items: one-pieces (saree, gown, dress) and top + bottom pairs.
3. Score	Each combination gets points for the occasion, style, colour harmony, weather and style profile.
4. Complete	Shoes, jewellery and a bag tagged for that occasion are added.
5. Choose and explain	Up to 4 looks that really differ are picked, each with reasons and honest cautions.

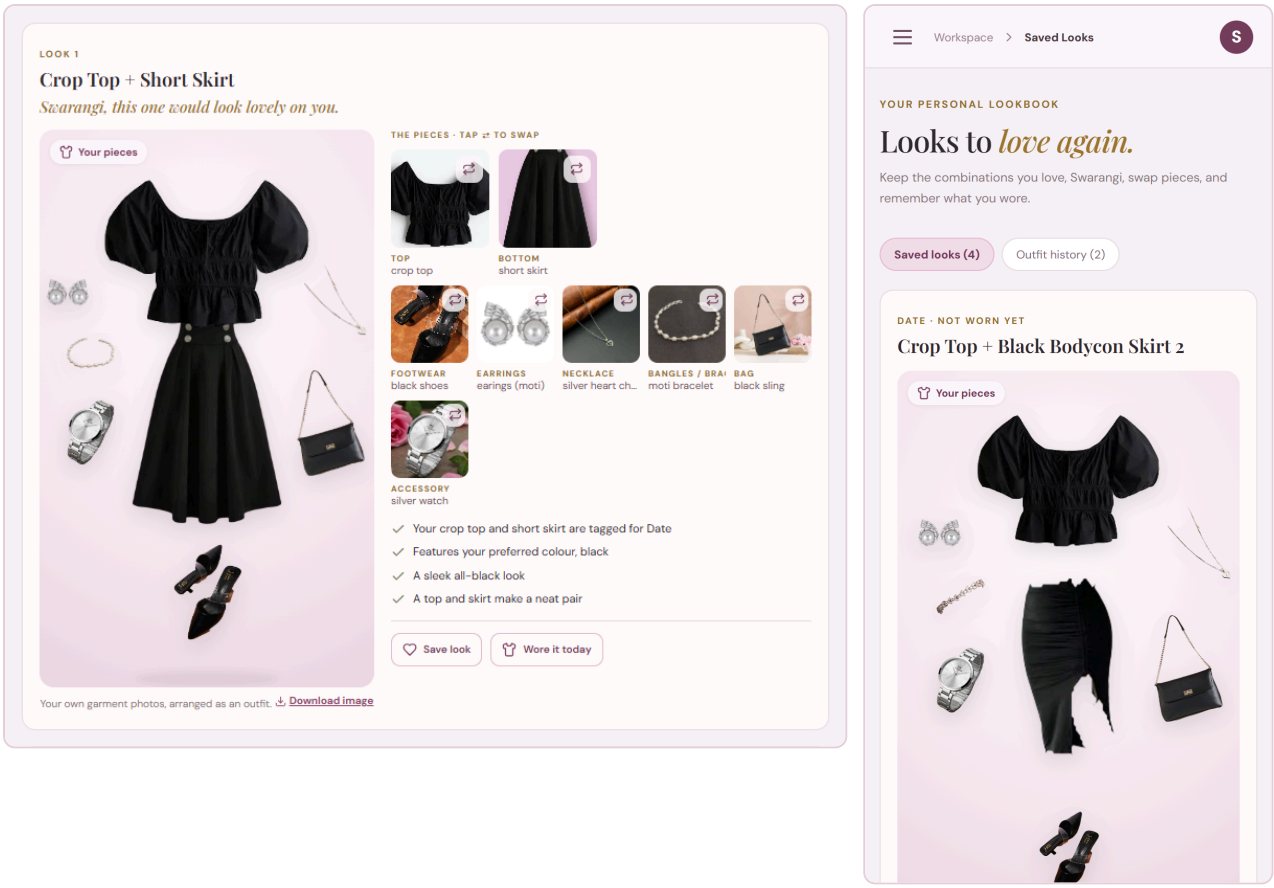


Figure 3. Real screenshots: an outfit made from the user's own pieces (left) and a saved look on a phone (right).

5 Technology Stack

Layer	Technology	In simple words
Frontend	React 18, Vite, plain CSS, Lucide icons	React builds the screens; Vite bundles them into fast files for the browser.
Backend	Python, FastAPI, Uvicorn, Pydantic	FastAPI answers the app's requests; Pydantic checks that incoming data is valid.
Database	SQLite (WAL mode)	A whole database in one file; WAL lets reading and writing happen together.
Styling logic	Rule-based Python	Hand-written styling rules instead of a paid AI service.
In-browser AI	onnxruntime-web, IS-Net, SegFormer-B2 clothes parser	Free AI models that run inside the browser to remove photo backgrounds and recognise clothes.
Weather	Open-Meteo API	Free weather service, no account or key needed.
Voice	Web Speech API	The browser's built-in speech-to-text.
Email	SMTP (Gmail app password)	Sends email-confirmation and password-reset links.
Hosting	Azure for Students VM, Ubuntu 24.04, Caddy, systemd	A free student server; Caddy provides https; systemd keeps Adaa running and restarts it if needed.
Backups	Python `backup.py`, Backblaze B2	Nightly copies kept on the server and off-site.
Monitoring	UptimeRobot	Checks the site every 5 minutes and emails if it goes down.
Testing	pytest (258 tests)	Automatic checks that the backend and styling rules still work.

6 Features

Feature	Purpose
My Wardrobe	Add, search, filter, edit and delete pieces, with photo auto-fill.
AI Stylist	Ask for outfits by typing, voice or options; get up to 4 explained looks.
Outfit boards	See each look as the user's own garment photos arranged as if worn; download as an image.
Swap, save, history	Change any piece, save looks, add a photo wearing a look, record what was worn.
Wedding Planner	A different look per function (haldi to reception), following dress codes, no garment repeated.
Festivals	About 30 festivals and national days with their colours; Navratri colour of the day.
Trip Planner	One outfit per trip day without repeating a garment, plus a packing list.
Weather-aware styling	Dresses the user for today's temperature and rain in their city.
Wear Stats	Pieces not worn lately, most worn, cost per wear.
Shop Smart	Which one new piece would unlock the most new outfits.
Style profile	Optional details that add "Why it suits you" notes to each look.

Accounts	Private wardrobe per person, email confirmation, password reset, delete account.
Home-screen app	Can be installed on a phone or laptop and opens like an app.

7 Feature-wise Implementation

Feature	Technology used	How it works
Wardrobe and photos	React form, FastAPI, SQLite, `storage.py`, `image_meta.py`	The browser shrinks the photo (max 1600 px). The server checks the real file type, saves it under a random name and removes hidden data like GPS location.
Auto-fill from photo	SegFormer-B2 clothes parser in a Web Worker	The model labels the garment's pixels (top, skirt, trousers…); the shape and main colour fill in the form. Only untouched fields are filled.
Background removal	IS-Net + clothes parser, onnxruntime-web	Runs once per garment in a background thread of the browser; the clean picture is saved on the server and reused.
Understanding requests	`recommender/parser.py`	Keyword rules with small typo tolerance; knows Hindi colour and occasion words (lal, shaadi, safar…).
Choosing outfits	`recommender/engine.py`, `colours.py`, `garments.py`	Builds every valid combination, scores it, adds accessories, picks up to 4 that differ, and writes reasons and cautions.
Voice requests	Web Speech API (Indian English)	Speech becomes text in the browser, then goes through the same parser.
Outfit board	`OutfitBoard.jsx`, `boardImage.js`	Places the cut-outs as if worn, sized to each other; "Download image" redraws it on a canvas as a PNG.
Wedding Planner	`recommender/events.py` + planner search	Each function has dress-code colours and colours to avoid; a search gives each function a different look without repeating any garment.
Trip Planner	`recommender/planner.py`	A search fills the days with the best outfits so no garment is worn twice; builds a packing list.
Festivals	`recommender/festivals.py`	A table of festivals, their spellings, colours and dates; requests for a festival show only outfits in its colours.
Weather	Open-Meteo, `weather.js`, `recommender/weather.py`	The browser fetches today's forecast for the saved city; hot, cold and rain gently change the scores and add notes.
Wear Stats	SQLite history tables, `planner.py`	Counts each "Wore it today"; works out days since worn and price ÷ wears.
Shop Smart	`planner.py`	Tries 17 versatile pieces against the wardrobe and counts how many new outfits each would make.
Style profile	`recommender/profile_fit.py`	Reads each piece's cut from its type and name, adds small score bonuses and "Why it suits you" notes. Never hides clothes.

Accounts and email	`auth.py`, `mailer.py`, SMTP	Passwords hashed with scrypt; session cookie; one-use links emailed for confirmation and reset.
Hosting and backups	Caddy, systemd, `backup.py`, `offsite.py`	One service runs Adaa; a nightly timer backs up the database and photos and uploads a copy to Backblaze B2.

8 Security

Area	Measure	Why it matters
Passwords	Only a salted *scrypt hash* is stored (a one-way scramble)	Even someone with the database can't read passwords.
Sign-in sessions	Random token in an HttpOnly, SameSite cookie (Secure on https); database keeps only its SHA-256	Scripts can't read the cookie, and a copied database can't be used to log in.
Guessing passwords	5 wrong tries per email in 15 minutes; per-address limits on sign-up, log-in, reset emails, uploads	Stops automated guessing and spam.
Private wardrobes	Every database query is filtered by the logged-in user	One account can never see or change another's data.
Email links	One-use, hashed, short-lived (30 min reset, 24 h confirm); placed after #	Links can't be reused and never appear in server logs.
Photos	Real type check, random names, size limit, hidden location data removed	Blocks disguised files and protects the user's location.
Web attacks	Cross-site write requests blocked; strict *Content Security Policy*; no third-party scripts or fonts	Prevents other sites from acting as the user and blocks injected scripts.
Connection	https everywhere (Caddy, Let's Encrypt), HSTS and other security headers	Data is encrypted between the phone and the server.
Server	Runs as its own limited user, may only write to its data folders; logs kept 30 days	Limits damage if something goes wrong.
Data loss	Nightly backups (7 kept) + off-site copy; tested restore	Data survives a server failure.
Your choices	Delete account any time (data and photos erased); privacy note and terms; 18+ only	Users stay in control of their data.

> No API keys or passwords are stored in the code. Server secrets live only in a settings file on the server, readable by the Adaa service alone.

9 Future Scope

Improvement	Benefit
Android app on Google Play	Easy to find and install (the web app wrapped as a Trusted Web Activity).
Own domain + transactional email	A memorable address, and emails that land in the inbox instead of spam.
Offline screen	A friendly message instead of a browser error when there's no internet.
Use wear history in styling	Rank pieces that haven't been worn lately higher.
Makeup suggestions	Complete the look beyond clothes and jewellery.

Optional local language model	Friendlier chat, while the rules still choose the clothes.
Commercially licensed clothes model	The current clothes parser's licence allows non-commercial use only.

Current limitations

Limitation	Detail
Keyword understanding	Clear sentences work best; Adaa always shows what it understood.
Auto-fill accuracy	On 28 real photos: category always right, type about 9 in 10, colour about 3 in 4.
Phones	Automatic background removal is skipped on phones (the model is about 200 MB).
Scale	One server and one SQLite file: suited to early users, not very large numbers.

All features, technologies and figures in this report come from the Adaa source code and its test records (docs/TESTING.md).